Association Between Urban Land Use and Pyrethroid Toxicity in California Waterways

Pandu Rupanagudi

October 12, 2026

ABSTRACT IS STILL INCOMPLETE

1 Introduction

Pyrethroids are synthetic insecticides widely used in agriculture and commercial pest-control products, including bug sprays, pet shampoos, and flea and tick treatments. They mimic pyrethrins, natural insecticidal compounds found in chrysanthemum flowers (Hołyńska-Iwan and Szewczyk-Golec 2020). Although primarily intended for pest control, pyrethroids can also harm non-target organisms, including bees, fish, and aquatic macroinvertebrates (Ranatunga, Kellar, and Pettigrove 2023). Since they are used in urban areas, pollutants from developed land may be transported into nearby waterways. Moreover, California's Stream Pollution Trends Monitoring Program (SPoT) found associations between pyrethroid concentrations, toxicity, and urban land use in California watersheds (Phillips et al. 2020). After learning this information, I ask the research question: is there an association between how toxic a waterway is in terms of pyrethroids and how much of the surround land is urban?

Hence, this study examines whether the percentage of urban land surrounding a monitoring station is associated with the sum of pyrethroid toxic units measured at that station. Using data from 100 California SPoT monitoring stations sampled in 2013, I fit a simple linear regression model with `PercentUrbanLandUse` as the explanatory variable and `SumPyrethroid_ToxicUnits` as the response variable (California State Water Resources Control Board 2019). Since the data are observational, the analysis evaluates association rather than causation.

Further, I also conduct a two-sided hypothesis test to determine whether there is a linear association between the `PercentUrbanLandUse` and `SumPyrethroid_ToxicUnits` variables. The hypothesis test reveals that there is no evidence for a linear association for between the two variables. This is further explained in Section 4.1.

The structure of this paper is as follows:

- Section 2 explains the dataset used and its relevant details
- Section 3 details the statistical methods used
- Section 4 describes the fitted model and its implication
- Section 5 discusses the key findings and its implications in regards to the research question

The code for this project is available on GitHub[1].

2 Data

2.1 Dataset background

I will be using the Sum Pyrethroid Toxic Units dataset provided by the California State Water Resources Control Board (2019). In this paper, the data will be referred to as the pyrethroid dataset for brevity. The pyrethroid dataset was complied for California's 2019 water quality status report by the Stream Pollution Trends (SPoT) Monitoring Program. The SPoT program was initiated to monitor and manage the contamination of streams statewide in California.

2.2 Definitions of key terms

The key terms to understanding this paper and its context are described below:

- **Waterway**: a navigable body of water such as a river, stream, or canal that may be used to follow along using a boat or other means of water transportation.
- **Watershed**: an area of land where all the precipitation collects into downhill river, stream or other waterway.
- **Pyrethroids**: man-made insecticides commonly used in farming and commercial pest-control products.

2.3 Variables summary

The dataset contains 807 observations and these 6 relevant columns:

`StationCode`: a unique code that identifies a monitoring watershed station in California.

`PercentUrbanLandUse`: the percentage of the surrounding land classified as urban in a 5-km radius of a SPoT monitoring watershed station.

[1]Link to the project repository: https://github.com/pandu-0/waterways-toxicity

`SumPyrethroid_ToxicUnits`: the potential for a pyrethroid pesticide to cause toxicity. The values can range from 0 to any positive real number.

`Year`: the year a sample of sediment was collected and tested for pyrethroid toxicity from a SPoT monitoring watershed.

`SampleDate`: the date a sample of sediment was collected and tested for pyrethroid toxicity from a SPoT monitoring watershed.

`LandUse`: the category that the surrounding land of a SPoT monitoring watershed is classified as (values can be "Urban", "Agriculture", "Open", "Ag/Urban").

Each row in this dataset represents a sample of sediment taken from the watershed station and tested for the potential for a pyrethroid pesticide to cause toxicity, which is the `SumPyrethroid_ToxicUnits` variable. Moreover, the dataset has samples from the 2008 to 2017. In order to eliminate this time dependence of samples in this dataset, I will only be using samples for the year 2013 to fit a linear regression model and answer the research question. This brings the dataset to a subset of 100 rows with all 100 unique SPoT monitoring stations that exist in California. I should also note the 2013 subset contains no missing values in either variable used in the regression.

The Figure 1 below shows locations of the watersheds in the dataset.

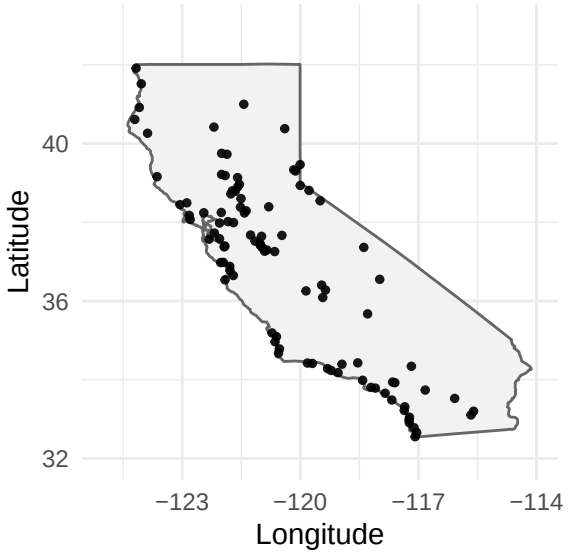


Figure 1: California Watershed SPoT Monitoring station locations

2.4 Key Variables

The key variables that are used to answer the research question are `PercentUrbanLandUse` and `SumPyrethroid_ToxicUnits`.

2.4.1 Percent Urban Land Use

`PercentUrbanLandUse` represents the percentage of land classified as urban within the 5-km drainage area surrounding each SPoT monitoring station. Researchers mapped these drainage areas using elevation data and extracted land-cover information from the National Land Cover Database (NLCD). Urban land cover included developed open space and low, medium, and high-intensity developed areas (NLCD categories 21–24) (Phillips et al. 2020).

2.4.2 Sum Pyrethroid Toxic Units

In the dataset chosen for this paper, `SumPyrethroid_ToxicUnits` represents the potential for a pyrethroid pesticide to cause toxicity, according to Phillips et al. (2020). In order to record these values at the various waterways in California, SPoT surveys are conducted every year between late spring to fall, where sediment is collected from recent stream bed deposits after a high flow season and tested for contaminants like pyrethroid pesticides. The SPoT 10 year report states that the contamination results should not be interpreted "as a characterization of the entire stream in which study sites were located" (Phillips et al. 2020). Interestingly, pyrethroids are hydrophobic so they do not dissolve in water (Laskowski 2002). They instead drop down to the bottom of the stream and collect as sediment, which is why the `SumPyrethroid_ToxicUnits` readings from the dataset cannot be used to characterize an entire stream's pyrethroid toxicity. The report rather suggests that the readings should be understood as relative indicators of annual pollutant mobilization and transport within the watersheds. In simple terms, the `SumPyrethroid_ToxicUnits` values are only an indicator of how much pollution is moved through the waterway. In the dataset, 40 values out of 100 for `SumPyrethroid_ToxicUnits` are 0. This most likely due to the fact these pyrethroid readings correspond to watersheds location at areas that are classified as Open or Agricultural for the their `LandUse`.

3 Method

To find whether there is an association between the `PercentUrbanLandUse` and `SumPyrethroid_ToxicUnits` in California waterways, I fit a simple linear regression model where the predictor variable is `PercentUrbanLandUse`, denoted as X_i, and the response variable is the `SumPyrethroid_ToxicUnits`, denoted as Y_i.

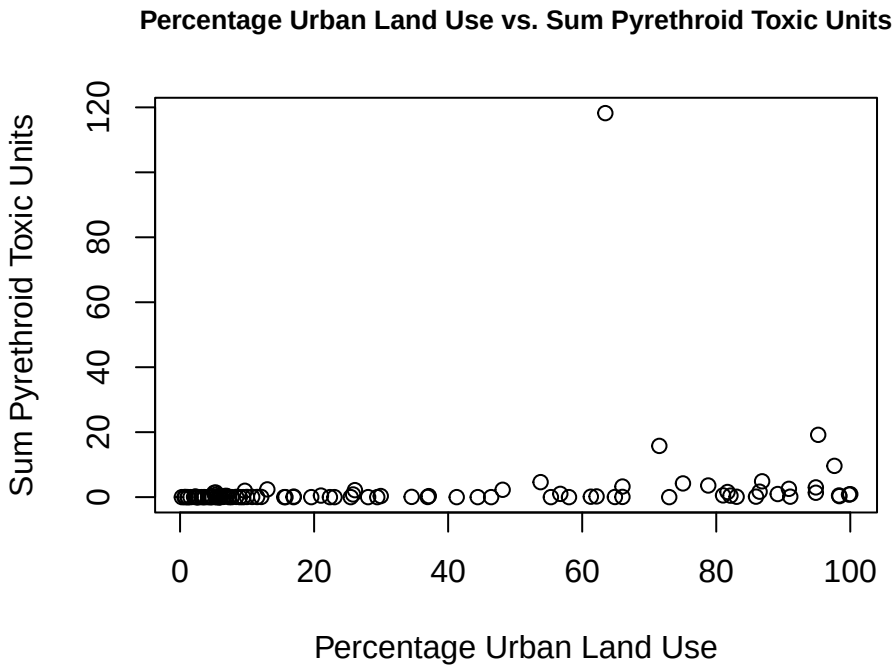


Figure 2: Scatter plot of Percent Urban Land use and Sum Pyrethroid Toxicity

The simple linear regression model can be written as follows for all observations $i = 1$ to n:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

β_0 is the intercept of the regression line. In the context of this paper, it represents the mean value of `SumPyrethroid_ToxicUnits` when `PercentUrbanLandUse` is 0.

β_1 is the slope of the regression line. β_1 represents the expected increase in Sum Pyrethroid toxicity for each unit increase in the percentage of urban land use.

The sampling distribution of $\frac{b_1 - \beta_1}{\widehat{se}(b_1)}$ is $t(n-2)$. I will use this fact to test whether there is an association between the percent of urban land use and Pyrethroid toxicity in California waterways using a hypothesis test with confidence intervals. However, in order to draw any conclusions from the hypothesis test we must first check if the assumptions for this method are met.

3.1 Linear model assumptions

The simple linear regression model assumes the following conditions:

1. The predictors and response variables accurately represent the quantities of interest and answer the research question

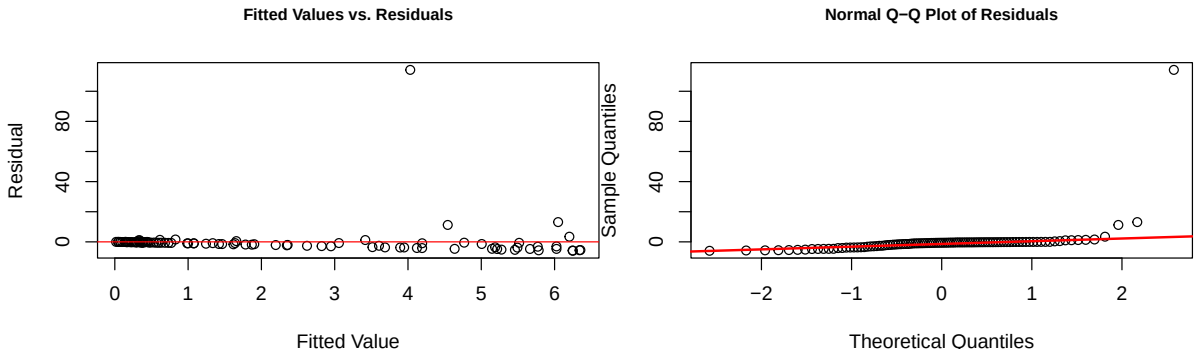
2. The dataset is representative of the population of interest
3. The mean function is correct
4. All errors ε_i are independent
5. Errors are normally distributed and have finite variance

3.2 Testing the assumptions

1. The predictor variable used in this model is the percentage of urban land use and the response variable is the Sum Pyrethroid toxicity units. Based on the dataset, the variables accurate represent the quantities such that the research question can be answered.
2. The stations are distributed across California as shown in Figure 1, but they were selected through a targeted monitoring design rather than random sampling. Therefore, the results should be generalized primarily to SPoT-monitored watersheds and interpreted cautiously when discussing California waterways more broadly.
3. The scatter plot comparing `PercentUrbanLandUse` and `SumPyrethroid_ToxicUnits` in Figure 2 and the residual plot in Figure 3a do not show any evidence of the relationship between the variables as non-linear. Therefore, there is no reason to conclude that the fitting a linear model for this case would be incorrect.
4. Knowing the sample information about one SPoT monitoring watershed could tell us information about surrounding watershed's `SumPyrethroid_ToxicUnits` values due to the fact that the watersheds chosen when compiling this dataset were not randomly chosen, they were instead picked for a targeted monitoring program. Hence, the errors may not be independent since samples collected from the same region in California could have similar `SumPyrethroid_ToxicUnits` values.
5. The residual plot below shows that there is unequal variance in the errors. There is also a very noticeable outlier in the data as well which can be seen in Figure 3b. Moreover, to check for normality of errors I have used a QQ-plot as seen in Figure 3b. Figure 3b shows that the error quantiles line up with the normal quantiles line with some deviation towards the right end. However, I will still assume that the errors are normal.

4 Results

I used the `lm()` function in R to fit a simple regression model (R Core Team 2025). The estimated slope parameter b_1 is 0.0635150. This means that the variable `SumPyrethroid_ToxicUnits` is expected to increase by 0.0635150 for each unit in increase of `PercentUrbanLandUse`. The estimated intercept parameter b_0 is 0.0008118. This means that when `PercentUrbanLandUse`



(a) Residual plot

(b) QQ-plot of the residuals

Figure 3: Residual plot shows unequal variance and QQ-plot shows errors are mostly normal with a visible outlier

is 0, the average `SumPyrethroid_ToxicUnits` is 0.0008118. The coefficient of determination, R^2 is 0.03223. This means that approximately 3.22% of the variability in `SumPyrethroid_ToxicUnits` is explained by the predictor `PercentUrbanLandUse`.

4.1 Hypothesis Test

To test whether there is an association between the `PercentUrbanLandUse` and `SumPyrethroid_ToxicUnits`, I use a two-sided hypothesis test with the null and alternative hypothesis as follows:

$$H_0 : \beta_1 = 0, H_a : \beta_1 \neq 0$$

In order to draw any meaningful conclusions from this hypothesis test, it is necessary that the requirements for this test are met: the errors must be independent and normal (Kutner et al. 2005). As noted in Section 3.2 that the errors are normal, however, they may not be independent. This puts a limitation on this experiment because violating the assumptions means that the hypothesis test is less reliable (Kutner et al. 2005). I will conduct the hypothesis test bearing this mind.

If we fail to reject H_0, then we did not find evidence that there is a linear association between the predictor and response variables. On the other hand, if we reject H_0 in favor of H_a, then we say that found evidence that there is a linear association between (Kutner et al. 2005).

Under the null hypothesis, the distribution of $\frac{b_1 - \beta_1}{\hat{se}(b_1)}$ is $t(n-2)$. Using this distribution, I will build a 95% confidence interval for β_1 with significance level $\alpha = 0.05$ to control risk of a Type I error. If this interval contains 0, then we fail to reject H_0. If this interval does not contain 0, then we reject H_0 in favor of H_a. The 95% confidence interval for β_1 is $b_1 \pm t_{1-\frac{\alpha}{2};\, n-2} \cdot \hat{se}(b_1)$.

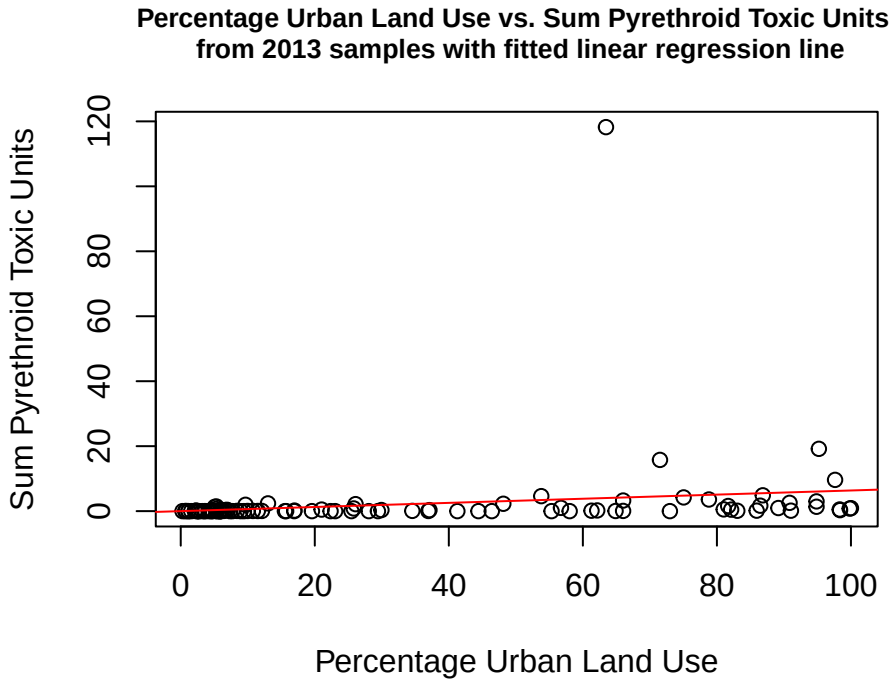


Figure 4

The estimate for b_1 is 0.0635150, standard error of b_1 is 0.0351555, and we have 100 observations in the dataset, so the interval can be constructed with the values as follows: $0.0635150 \pm t_{0.975;\,98} \cdot 0.0351555$. Once simplified the confidence interval approximately becomes $-0.0062 \leq \beta_1 \leq 0.1333$. Since this interval contains 0, we fail to reject H_0. If the conditions of the test are met, then we can say that we did not find evidence that there is a linear association between the predictor and response variables.

5 Discussion

References

California State Water Resources Control Board. 2019. "Fig. 4-5. Sum Pyrethroid Toxic Units." California Open Data. https://data.ca.gov/dataset/2019-california-water-quality-status-report/resource/4f9dcf98-5fb4-477b-8780-4411c083fee8.

Hołyńska-Iwan, Iga, and Karolina Szewczyk-Golec. 2020. "Pyrethroids: How They Affect Human and Animal Health?" *Medicina* 56 (11): 582. https://doi.org/10.3390/medicina56110582.

Kutner, Michael H., Christopher J. Nachtsheim, John Neter, and William Li. 2005. *Applied Linear Statistical Models*. 5th ed. Boston, MA: McGraw-Hill/Irwin.

Laskowski, Dennis A. 2002. "Physical and Chemical Properties of Pyrethroids." In *Reviews of Environmental Contamination and Toxicology*, 174:49–170. Springer. https://doi.org/10.1007/978-1-4757-4260-2_3.

Phillips, Bryn M., Katie Siegler, Jennifer P. Voorhees, Laura McCalla, Sylvia Zamudio, Katie Faulkenberry, Ali Dunn, Tessa Fojut, and Brian Ogg. 2020. "Spatial and Temporal Trends in Chemical Contamination and Toxicity Relative to Land Use in California Watersheds: Stream Pollution Trends (SPoT) Monitoring Program." Fifth Report. Sacramento, CA: California State Water Resources Control Board.https://www.waterboards.ca.gov/water_issues/programs/swamp/spot/docs/spot_10_year_report.pdf .

R Core Team. 2025. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. https://www.R-project.org/.

Ranatunga, Madara, Claudette Kellar, and Vincent Pettigrove. 2023. "Toxicological Impacts of Synthetic Pyrethroids on Non-Target Aquatic Organisms: A Review." *Environmental Advances* 12: 100388. https://doi.org/https://doi.org/10.1016/j.envadv.2023.100388.